



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 5: Consumer Theory III

Patricia Galilea A.

ICS2523 Microeconomics

Ayudante de Bienestar

Diego Aguilera

Transporte y MPP

diego.aguilera@uc.cl

+56 9 8748 4067



Lecture 5

Last lecture we

- Learnt about the Budget constraint
- Understood consumer's choice
- Learnt a **mathematical approach** to model consumer's choice

In today's lecture we will

- Learn about the **Marshallian Demand**
- Learn the difference between **Marshallian and Hicksian Demand**
- Review the duality in the consumer theory
- Learn some important identities

Example: Cobb-Douglas (L4)

An individual has a Cobb-Douglas utility function given by $U(x, y) = x^\alpha y^\beta$, where $\alpha + \beta = 1$

And has a budget constraint given by

$$P_x x + P_y y = I$$

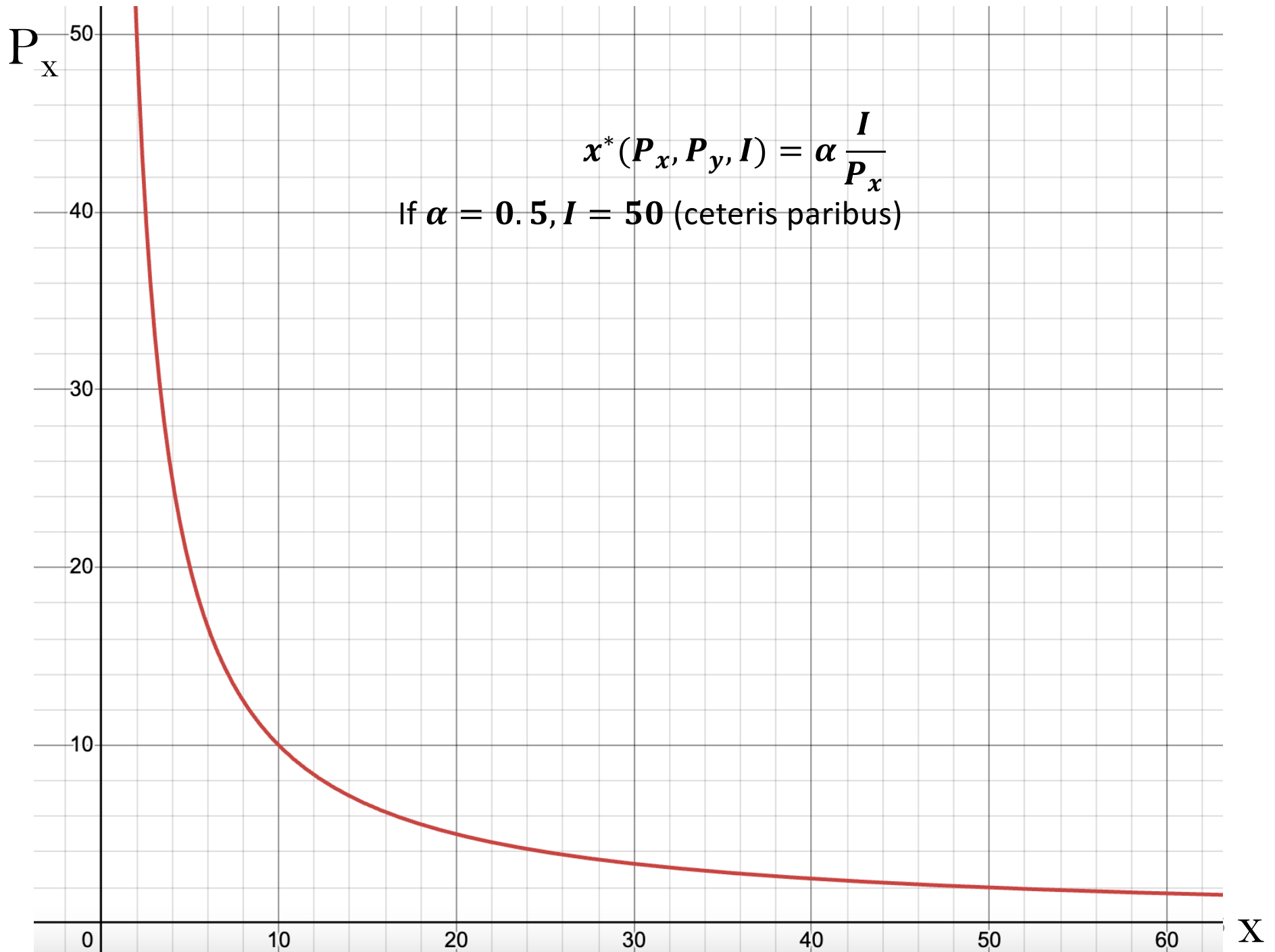
Solve the utility maximizing values for x and y .

The optimal values are:

$$x^*(P_x, P_y, I) = I \frac{\alpha}{\alpha + \beta} \frac{1}{P_x} = \alpha \frac{I}{P_x}$$

$$y^*(P_x, P_y, I) = \beta \frac{I}{P_y}$$

Demand
of x and
 y , given
 P_x, P_y, I



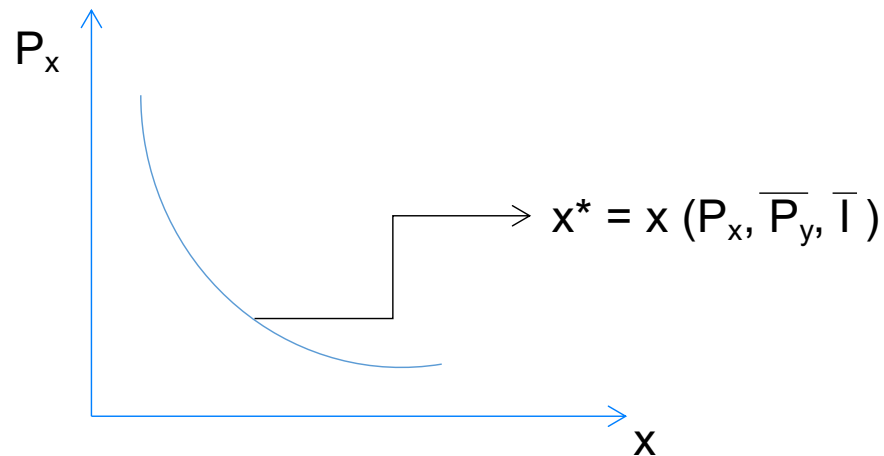
Marshallian Demand

Consumer Theory

5. Marshallian Demand

Marshallian demand: The solution to the problem of utility maximization subject to the budget constraint

$$x_i^* = x_i(P_1, P_2, \dots, P_n, I)$$



Remember: ceteris paribus

5. Marshallian Demand

Marshallian demand is more widely used because the data needed to estimate them (prices and nominal incomes) are readily available. They are also called *uncompensated demand* curves.

Marshallian demand function is *homogeneous of degree zero* in Prices and Income →

$$x(tP_x, tP_y, tI) = t^0 x(P_x, P_y, I)$$

Activity 1

An individual has a Cobb-Douglas utility function given by

$$U(x, y) = x^{\alpha} y^{\beta}, \text{ where } \alpha + \beta = 1$$

And has a budget constraint given by

$$P_x x + P_y y = I$$

Show that its Marshallian demand function is
homogeneous of degree zero in Prices and Income

5. Marshallian Demand

Indirect utility function ($V(P_1, \dots, P_n, I)$): This function gives us the maximum utility achievable at given prices and income.

If we solve the utility maximization problem subject to income restriction and replace the value of x^* and y^* into the utility function, we will get $V(P_1, \dots, P_n, I)$

$$U^* = U(X_1^*, X_2^*, \dots, X_n^*) = U(X_1(P_1, P_2, \dots, P_n, I), \dots, X_n(P_1, P_2, \dots, P_n, I)) = V(P_1, \dots, P_n, I)$$

Why is it called **indirect**?

5. Marshallian Demand

Indirect utility function: Example

In the previous example for Cobb-Douglas: $U(x, y) = x^\alpha y^\beta$, where $\alpha + \beta = 1$

And $\mathbf{x}^* = I\alpha/P_x$, $y^* = I\beta/P_y$

$$U(x, y) = x^\alpha y^\beta$$

$$V(P_x, P_y, I) = \left(\frac{\alpha I}{P_x}\right)^\alpha \left(\frac{\beta I}{P_y}\right)^\beta = I \frac{\alpha^\alpha \beta^\beta}{P_x^\alpha P_y^\beta}$$

5. Marshallian Demand

Indirect utility function

$V(P_1, \dots, P_n, I)$ is **continuous** in prices and income

$V(P_1, \dots, P_n, I)$ is **non-increasing** in prices and strictly increasing in I

$V(P_1, \dots, P_n, I)$ is **convex** in prices

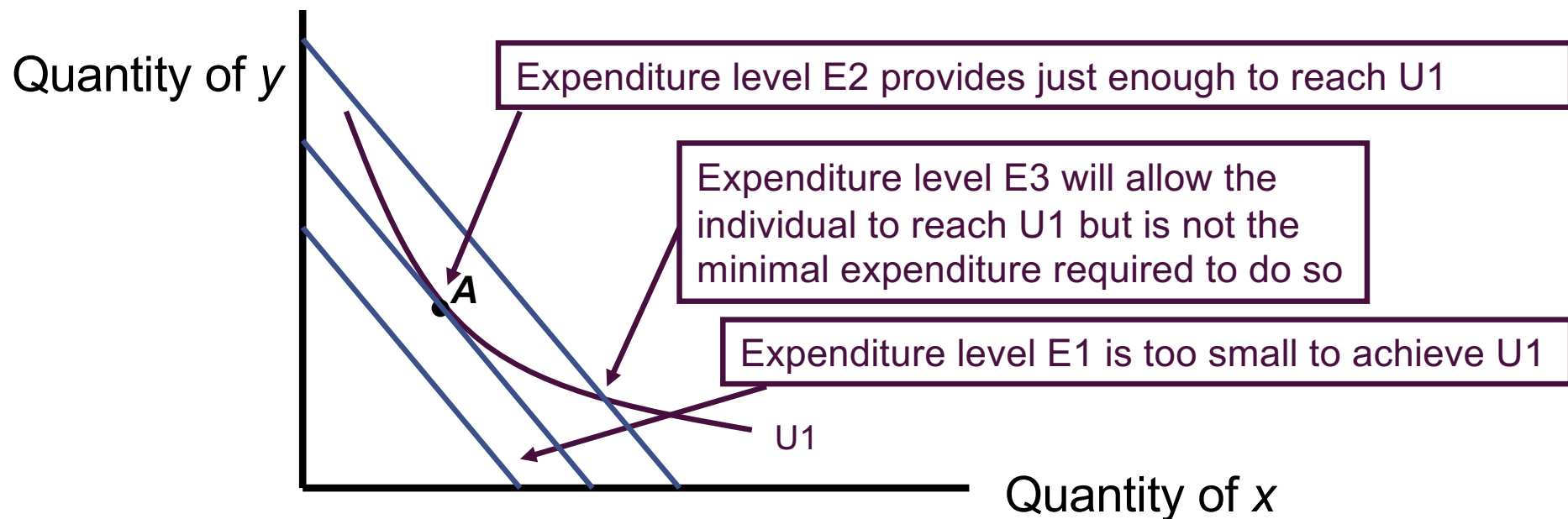
$V(P_1, \dots, P_n, I)$ is **homogeneous of degree zero** in prices and income

Duality in Consumer Theory

Consumer Theory

Duality in Consumer Theory

Instead of maximizing individual's utility subject to his/her income, we can also minimize his/her expenditure subject to his/her level of utility



Duality in Consumer Theory

Hicksian demand (compensated demand function): Tell us what consumption bundle achieves a target level of utility and minimizes total expenditure

$$h_i^* = h_i(P_1, P_2, \dots, P_n, U)$$

Hicksian demand function is *homogeneous of degree zero* in Prices

Activity 2

An individual has a Cobb-Douglas utility function given by

$$U(x, y) = x^{\alpha} y^{\beta}, \text{ where } \alpha + \beta = 1$$

And has a budget constraint given by

$$P_x x + P_y y = I$$

Find its Hicksian demand function and show that it is homogeneous of degree zero in Prices.



Author: Michael Hund (2023)

Duality in Consumer Theory

Expenditure function $E(P_1, P_2, \dots, P_n, U)$: Is the minimal amount of income necessary to achieve utility U at prices $P_1, P_2, \dots, P_n$.

$$E^* = P_1 X_1^* + P_2 X_2^* + \dots + P_n X_n^* = E(P_1, P_2, \dots, P_n, U)$$

Duality in Consumer Theory

Properties of the expenditure function:

- $E(P_1, P_2, \dots, P_n, U)$ is **continuous** in prices
 - $E(P_1, P_2, \dots, P_n, U)$ is **non-decreasing** in prices
 - $E(P_1, P_2, \dots, P_n, U)$ is **concave** in prices
 - $E(P_1, P_2, \dots, P_n, U)$ is **homogeneous degree one** in prices
-
- If h_i^* is the expenditure minimizing bundle necessary to achieve utility level U at prices $(P_1, P_2, \dots, P_n)$, then:

$$h_i^* = \frac{\partial E(P_1, \dots, P_n, U)}{\partial P_i}$$

for $i=1, \dots, n$ assuming the derivative exists and that $P_i > 0$

Activity 3

Camila consumes only cookies (c) and books (b). Her monthly income is I . The prices for the books and cookies are P_b and P_c , respectively. She has a quasilinear utility function given by:

$$U(c, b) = \ln(c) + b, \text{ where } P_b < I$$

And has a budget constraint given by

$$P_c c + P_b b = I$$

- a) Find its Marshallian demand and the indirect utility function.
- b) Find its Hicksian demand function and the expenditure function.

Important Identities

1. Roy's Identity:

$$x_i^* = x_i(P_1, \dots, P_n, I) \equiv - \frac{\frac{\partial V(P_1, \dots, P_n, I)}{\partial P_i}}{\frac{\partial V(P_1, \dots, P_n, I)}{\partial I}}$$

2. Shepard's lemma:

$$h_i^* = h_i(P_1, \dots, P_n, U) \equiv \frac{\partial E(P_1, \dots, P_n, U)}{\partial P_i}$$

$$3. \quad V(P_1, \dots, P_n, E(P_1, \dots, P_n, U)) \equiv U$$

Important Identities

$$4. \quad E(P_1, \dots, P_n, V(P_1, \dots, P_n, I)) \equiv I$$

$$5. \quad x_i(P_1, \dots, P_n, I) \equiv h_i(P_1, \dots, P_n, V(P_1, \dots, P_n, I))$$

$$6. \quad h_i(P_1, \dots, P_n, U) \equiv x_i(P_1, \dots, P_n, E(P_1, \dots, P_n, U))$$

Example

The Cobb-Douglas function, perfect substitutes and perfect complements are special cases of the more general constant elasticity of substitution function (CES), which takes the form

$$U(x, y) = \frac{x^\delta}{\delta} + \frac{y^\delta}{\delta}$$

Where $\delta \leq 1, \delta \neq 0$ and $U(x, y) = \ln x + \ln y$ when $\delta = 0$

For an individual with a CES utility function ($\delta \neq 0$) find his/hers:

- Marshallian demand function
- Indirect utility function
- Hicksian demand function
- Expenditure function
- Demonstrate that Roy's identity and Shephard's lemma

Example: solution

a) Marshallian demand function

$$\begin{aligned} \text{Max } U(x, y) &= \frac{x^\delta}{\delta} + \frac{y^\delta}{\delta} \\ \text{Subject to: } P_x x + P_y y &= I \end{aligned}$$

Where $\delta \leq 1, \delta \neq 0$ and $U(x, y) = \ln x + \ln y$ when $\delta = 0$

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

$$\frac{\cancel{\delta} x^{\delta-1}}{\cancel{\delta}} = \frac{\cancel{\delta} y^{\delta-1}}{\cancel{\delta}}$$

$$x^{\delta-1} = \frac{P_x}{P_y} y^{\delta-1},$$

$$x = \left(\frac{P_x}{P_y}\right)^{\frac{1}{\delta-1}} \cdot y \quad (*)$$

Example: solution

Replacing (*) in the budget constraint:

$$P_x x + P_y y = I$$

$$P_x \cdot \left(\frac{P_x}{P_y}\right)^{\frac{1}{\delta-1}} y + P_y y = I$$

$$\left(\frac{P_x^{\delta/(\delta-1)}}{P_y^{1/(\delta-1)}} + P_y \right) y = I$$

$$y_m^* = I \frac{P_y^{1/(\delta-1)}}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})}$$

Example: solution

Similarly:

$$x_m^* = I \frac{P_x^{1/(\delta-1)}}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})}$$

b) Indirect utility function

Replacing $x_m^*(P_x, P_y, I)$ and $y_m^*(P_x, P_y, I)$ in the utility function:

$$V(P_x, P_y, I) = \frac{(x_m^*)^\delta}{\delta} + \frac{(y_m^*)^\delta}{\delta}$$

Example: solution

b) Indirect utility function

$$V(P_x, P_y, I) = \frac{(x_m^*)^\delta}{\delta} + \frac{(y_m^*)^\delta}{\delta}$$

$$V(P_x, P_y, I) = I^\delta \frac{P_x^{\delta/(\delta-1)}}{\delta \cdot (P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^\delta} + I^\delta \frac{P_y^{\delta/(\delta-1)}}{\delta \cdot (P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^\delta}$$

$$V(P_x, P_y, I) = \frac{I^\delta \cancel{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^1}}{\delta \cdot (P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^{\delta - \delta - 1}}$$

$$V(P_x, P_y, I) = \frac{I^\delta}{\delta} \frac{1}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^{\delta-1}}$$

Example: solution

c) Hicksian demand function

$$\text{Min } P_x x + P_y y$$

$$\text{Subject to: } U(x, y) = \frac{x^\delta}{\delta} + \frac{y^\delta}{\delta} = \bar{U}$$

Similarly to a)

$$x = \left(\frac{P_x}{P_y}\right)^{\frac{1}{\delta-1}} \cdot y \quad (*)$$

Replacing (*) in the restriction:

$$\frac{\left[\left(\frac{P_x}{P_y}\right)^{\frac{1}{\delta-1}} \cdot y\right]^\delta}{\delta} + \frac{y^\delta}{\delta} = \bar{U}$$

Example: solution

$$\frac{\left[\left(\frac{P_x}{P_y}\right)^{\frac{1}{\delta-1}} \cdot y\right]^\delta}{\delta} + \frac{y^\delta}{\delta} = \bar{U}$$

$$\frac{\left(\frac{P_x}{P_y}\right)^{\frac{\delta}{\delta-1}} \cdot y^\delta}{\delta} + \frac{y^\delta}{\delta} = \bar{U}$$

$$y^\delta = \frac{\bar{U}\delta \cdot P_y^{\delta/\delta-1}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})}$$

Mental note:

$$1 + \frac{1}{\delta - 1} = \frac{\delta}{\delta - 1}$$

$$y_h^* = \frac{(\bar{U}\delta)^{1/\delta} \cdot P_y^{1/\delta-1}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{1/\delta}}$$

Example: solution

Similarly,

$$x_h^* = \frac{(\bar{U}\delta)^{1/\delta} \cdot P_x^{1/\delta-1}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{1/\delta}}$$

d) Expenditure function:

$$E(P_x, P_y, \bar{U}) = P_x \cdot x_h^* + P_y \cdot y_h^*$$

Mental note:

$$1 + \frac{1}{\delta - 1} = \frac{\delta}{\delta - 1}$$

$$= \frac{(\bar{U}\delta)^{1/\delta}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{1/\delta}} \cdot [P_x \cdot P_x^{1/\delta-1} + P_y \cdot P_y^{1/\delta-1}]$$

$$= \frac{(\bar{U}\delta)^{1/\delta}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{1/\delta}} \cdot [P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1}] \cdot \frac{1 - \delta}{\delta}$$

Mental note:

$$\frac{1}{\delta} - 1 = \frac{1 - \delta}{\delta}$$

Example: solution

d) Expenditure function:

$$E(P_x, P_y, \bar{U}) = \frac{(\bar{U}\delta)^{1/\delta}}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^{\frac{1-\delta}{\delta}}}$$

e) Roy's identity:

$$x_m^* = - \frac{\frac{\partial V(P_x, P_y, I)}{\partial P_x}}{\frac{\partial V(P_x, P_y, I)}{\partial I}}$$

$$V(P_x, P_y, I) = \frac{I^\delta}{\delta} \frac{1}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^{\delta-1}}$$

$$x_m^* = \frac{\frac{1}{\cancel{\delta}} \frac{I^\delta}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^\delta} \cdot (+ (\cancel{\delta-1}) \cdot \cancel{\frac{1}{\delta}} / (\cancel{\delta-1}) \cdot P_x^{\delta/(\delta-1)-1}}{\frac{1}{\cancel{\delta}} \frac{I^{\delta-1}}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^{\delta-1}}}$$

Mental note:
 $\frac{\delta}{\delta-1} - 1 = \frac{1}{\delta-1}$

Example: solution

e) Roy's identity:

$$x_m^* = - \frac{\frac{\partial V(P_x, P_y, I)}{\partial P_x}}{\frac{\partial V(P_x, P_y, I)}{\partial I}}$$

$$x_m^* = \frac{I^{\delta}}{I^{\delta-1}} \cdot \frac{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^{\delta-1}}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^{\delta}} \cdot P_x^{1/(\delta-1)}$$

$$x_m^* = I \frac{P_x^{1/(\delta-1)}}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})}$$

Example: solution

e) Shephard's Lemma

$$x_h^* = \frac{\partial E(P_x, P_y, \bar{U})}{\partial P_x}$$

$$E(P_x, P_y, \bar{U}) = \frac{(\bar{U}\delta)^{1/\delta}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{\frac{1-\delta}{\delta}}}$$

Mental note:

$$\frac{\delta}{\delta-1} - 1 = \frac{1}{\delta-1}$$

$$x_h^* = (\bar{U}\delta)^{1/\delta} \cdot \left(-\frac{1-\delta}{\delta}\right) \cdot \frac{1}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{\frac{1-\delta}{\delta}+1}} \cdot \frac{\delta}{\delta-1} \cdot P_x^{\frac{\delta}{\delta-1}-1}$$

$$x_h^* = (\bar{U}\delta)^{1/\delta} \cdot \frac{P_x^{\frac{1}{\delta-1}}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{\frac{1}{\delta}}}$$

Mental note:

$$\frac{1-\delta}{\delta} + 1 = \frac{1}{\delta}$$

Individual Activity

In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.

Activity 2: Solution

Now, we are minimizing expenditure subject to a certain level of Utility.

We already know that it is an interior solution, so we will go through the solution mathematically.

$$\text{Min}_{x,y} P_x x + P_y y$$

Subject to:

$$x^\alpha y^\beta = \bar{U} (\mu)$$

$$\mathcal{L}(x, y, \mu) = P_x x + P_y y + \mu(\bar{U} - x^\alpha y^\beta)$$

Activity 2: Solution

$$\mathcal{L}(x, y, \mu) = P_x x + P_y y + \mu(\bar{U} - x^\alpha y^\beta)$$

First order conditions:

$$\frac{\partial \mathcal{L}}{\partial x} = P_x - \mu \alpha x^{\alpha-1} y^\beta = 0 \rightarrow \mu = \frac{P_x}{\alpha x^{\alpha-1} y^\beta} \text{ (eq 1)}$$

$$\frac{\partial \mathcal{L}}{\partial y} = P_y - \mu \beta x^\alpha y^{\beta-1} = 0 \rightarrow \mu = \frac{P_y}{\beta x^\alpha y^{\beta-1}} \text{ (eq 2)}$$

$$\frac{\partial \mathcal{L}}{\partial \mu} = \bar{U} - x^\alpha y^\beta = 0 \text{ (eq 3)}$$

Activity 2: Solution

From eq.1 and eq. 2, we have

$$\frac{P_y}{\beta x^\alpha y^{\beta-1}} = \frac{P_x}{\alpha x^{\alpha-1} y^\beta}$$
$$x = y \frac{P_y \alpha}{P_x \beta}$$

In eq. 3 yields

$$\left(y \frac{P_y \alpha}{P_x \beta} \right)^\alpha y^\beta = \bar{U}$$

$$y_h = \bar{U} \left(\frac{P_x \beta}{P_y \alpha} \right)^\alpha, x_h = \bar{U} \left(\frac{P_y \alpha}{P_x \beta} \right)^\beta$$

Activity 2: Solution

$$y_h = \bar{U} \left(\frac{P_x \beta}{P_y \alpha} \right)^\alpha, x_h = \bar{U} \left(\frac{P_y \alpha}{P_x \beta} \right)^\beta$$

To test if these hicksian demands are homogenous of degree zero in Prices:

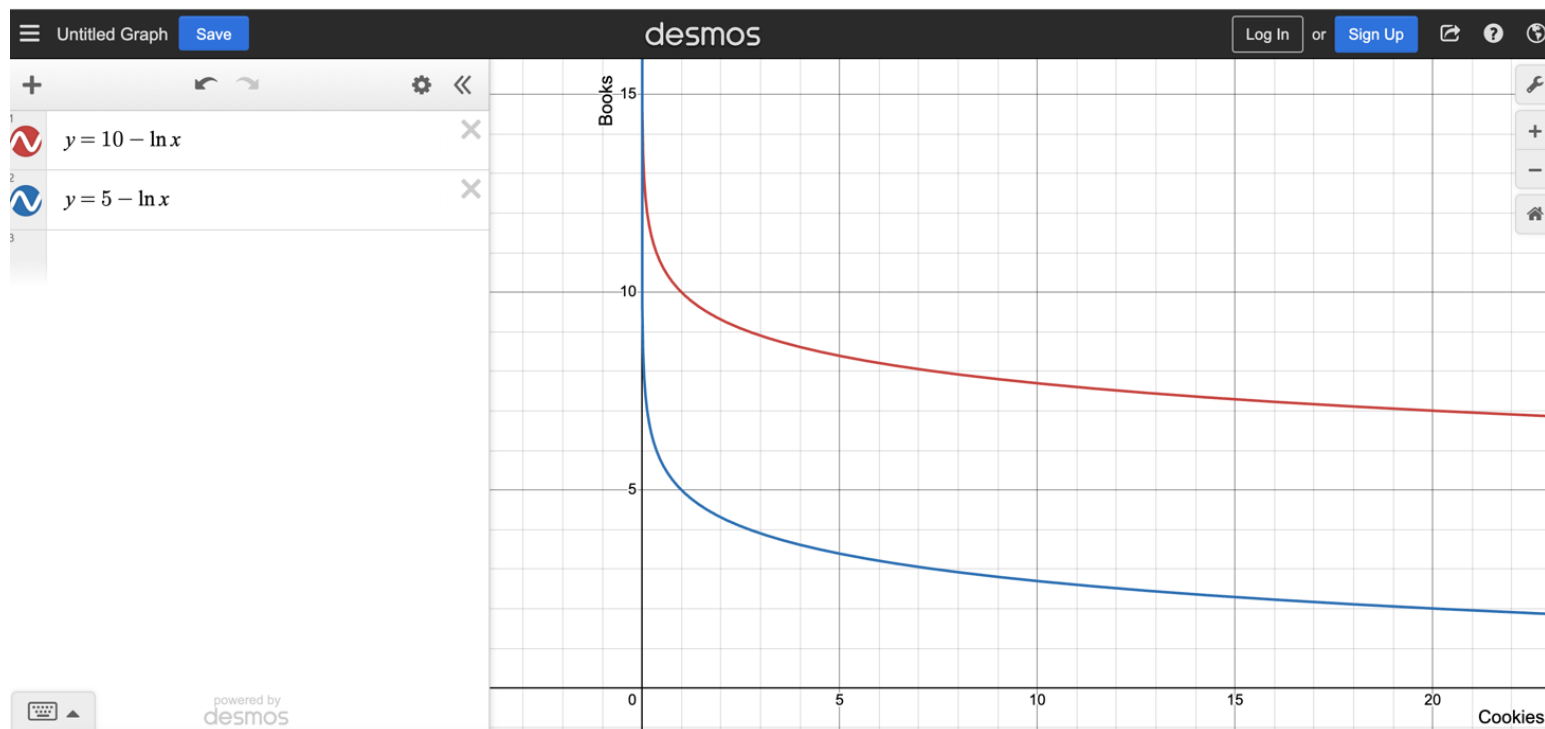
$$y_h = \bar{U} \left(\frac{tP_x \beta}{tP_y \alpha} \right)^\alpha = \bar{U} t^0 \left(\frac{P_x \beta}{P_y \alpha} \right)^\alpha$$

$$x_h = \bar{U} \left(\frac{tP_y \alpha}{tP_x \beta} \right)^\beta = \bar{U} t^0 \left(\frac{P_y \alpha}{P_x \beta} \right)^\beta$$

Activity 3: Solution

- a) Find its Marshallian demand and the indirect utility function.

$$U(c, b) = \ln(c) + b$$



Activity 3: Solution

- a) Find its Marshallian demand and the indirect utility function.

$$U(c, b) = \ln(c) + b$$
$$P_c c + P_b b = I$$

Using the first-order condition:

$$\frac{MU_c}{MU_b} = \frac{P_c}{P_b}$$

$$\frac{1/c}{1} = \frac{P_c}{P_b}$$

$$\frac{1}{c} = \frac{P_c}{P_b}$$

Activity 3: Solution

- a) Find its Marshallian demand and the indirect utility function.

$$\frac{1}{c} = \frac{P_c}{P_b}$$
$$c_m^* = \frac{P_b}{P_c} (*)$$

Replacing (*) in the budget constraint:

$$P_c c_m^* + P_b b = I$$
$$b_m^* = \frac{I}{P_b} - 1$$

Activity 3: Solution

- a) Find its Marshallian demand and the indirect utility function.

Given that $P_b < I$, the Marshallian demands are:

$$c_m^* = \frac{P_b}{P_c}$$
$$b_m^* = \frac{I}{P_b} - 1$$

(Note if $P_b \geq I$, b_m^* would be negative and that means that we are in a corner solution $\rightarrow b_m^* = 0$)

Activity 3: Solution

- a) Find its Marshallian demand and the indirect utility function.

Indirect Utility function:

$$V(P_c, P_b, I) = \ln(c_m^*) + b_m^*$$
$$V(P_c, P_b, I) = \ln\left(\frac{P_b}{P_c}\right) + \frac{I}{P_b} - 1$$

Let's check that $V(P_c, P_b, I)$ is homogenous of degree zero in prices and income:

$$V(\textcolor{red}{t}P_c, \textcolor{red}{t}P_b, \textcolor{red}{t}I) = \ln\left(\frac{\textcolor{red}{t}P_b}{\textcolor{red}{t}P_c}\right) + \frac{\textcolor{red}{t}I}{\textcolor{red}{t}P_b} - 1$$
$$= \ln\left(\frac{P_b}{P_c}\right) + \frac{I}{P_b} - 1 = \textcolor{red}{t}^0 V(P_c, P_b, I)$$

Activity 3: Solution

- b) Find its Hicksian demand function and the expenditure function.

$$\text{Min } P_c c + P_b b$$

Subject to:

$$\ln(c) + b = U$$

Using the same first-order conditions:

$$\frac{MU_c}{MU_b} = \frac{P_c}{P_b}$$

$$\frac{1/c}{1} = \frac{P_c}{P_b}$$

$$c_h^* = \frac{P_b}{P_c} (*)$$

Activity 3: Solution

- b) Find its Hicksian demand function and the expenditure function.

Replacing (*) in the utility level:

$$c_h^* = \frac{P_b}{P_c} (*)$$
$$\ln(c_h^*) + b = U$$

$$\ln\left(\frac{P_b}{P_c}\right) + b = U$$

$$b_h^* = U - \ln\left(\frac{P_b}{P_c}\right)$$

Activity 3: Solution

- b) Find its Hicksian demand function and the expenditure function.
These Hicksian demands are homogenous of degree zero in prices

$$c_h^* = \frac{P_b}{P_c}$$
$$c_h(\textcolor{red}{t}P_b, \textcolor{red}{t}P_c, U_0) = \frac{\textcolor{red}{t}P_b}{\textcolor{red}{t}P_c} = \textcolor{red}{t}^0 c_h(P_b, P_c, U_0)$$

$$b_h^* = U - \ln\left(\frac{P_b}{P_c}\right)$$
$$b_h(\textcolor{red}{t}P_b, \textcolor{red}{t}P_c, U_0) = U_0 - \ln\left(\frac{\textcolor{red}{t}P_b}{\textcolor{red}{t}P_c}\right) = \textcolor{red}{t}^0 b_h(P_b, P_c, U_0)$$

Activity 3: Solution

- b) Find its Hicksian demand function and the expenditure function.

Expenditure function:

$$E(P_c, P_b, U) = P_c c_h^* + P_b b_h^*$$

$$E(P_c, P_b, U) = P_c \frac{P_b}{P_c} + P_b (U - \ln \left(\frac{P_b}{P_c} \right))$$

$$E(P_c, P_b, U) = P_b + P_b (U - \ln \left(\frac{P_b}{P_c} \right))$$

This expenditure function is homogenous of degree one in prices:

$$\begin{aligned} E(\mathbf{t}P_b, \mathbf{t}P_c, U_0) &= \mathbf{t}P_b + \mathbf{t}P_b (U - \ln \left(\frac{\mathbf{t}P_b}{\mathbf{t}P_c} \right)) \\ &= \mathbf{t}^1 E(P_b, P_c, U_0) \end{aligned}$$